

Practical Session: PanGenie

EMBO Practical Course 2026, Naples



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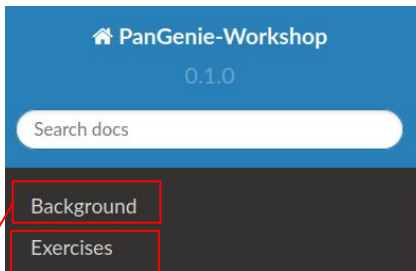
Heinrich Heine Universität Düsseldorf, Germany

PanGenie workshop

The workshop material is available here:

<https://pangenie-workshop.readthedocs.io/>

Solutions will be made available there after the workshop



🏠 / PanGenie workshop

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PanGenie workshop

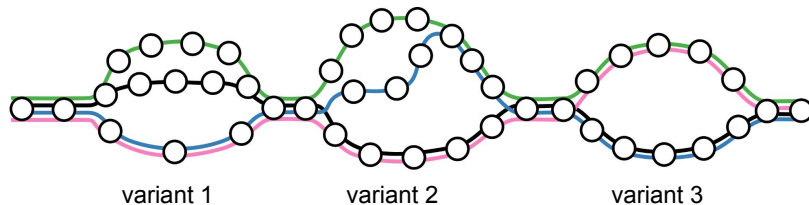
This workshop teaches the basics of PanGenie on a small use case. We provide background

Exercises we will be working on today

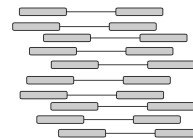
Details on PanGenie input data and commands

Input:

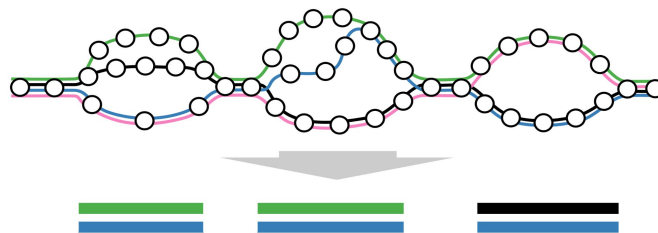
1. **Pangenome graph** constructed from known haplotypes



2. **Short reads** of the sample to be genotyped
(not in graph!)



Goal: Determine genotype of unknown genome for all variants contained in graph

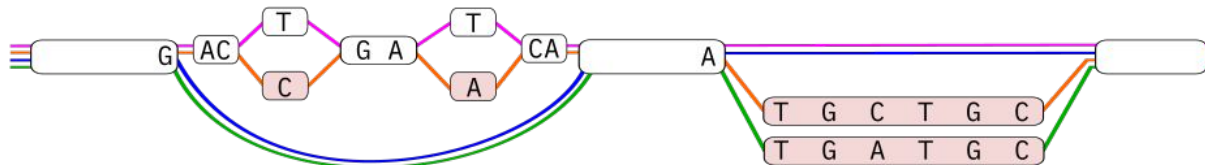


- **Reads**
 - WGS Illumina sequencing reads of a diploid individual
 - **FASTA** or **FASTQ** format (unaligned)
- **Reference genome**
 - reference genome in **FASTA** format
- **Pangenome reference**
 - pangenome reference represented in terms of a **VCF** file

Pangenome reference

- graphs build from haplotype-resolved assemblies
- variants produce **bubbles** in graph
- haplotypes represented as **paths**

pangenome graph



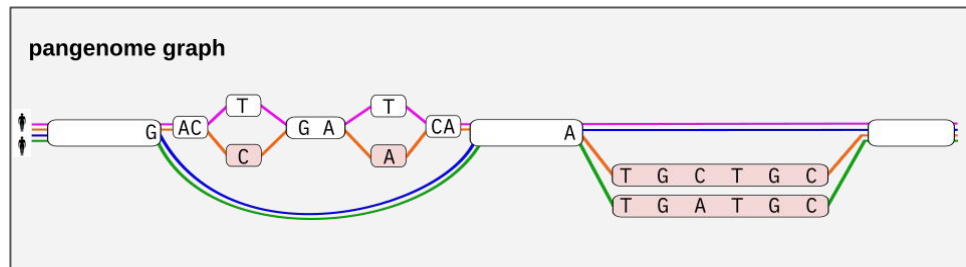
We use VCF format to encode the pangenome

```
##fileformat=VCFv4.2
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT NA00001 NA00002 NA00003
20 14370 rs6054257 G A 29 PASS NS=3;DP=14;AF=0.5;DB;H2 GT:GQ:DP:HQ 0|0:48:1:51,51 1|0:48:8:51,51 1/1:43:5:.,.
20 17330 . T A 3 q10 NS=3;DP=11;AF=0.017 GT:GQ:DP:HQ 0|0:49:3:58,50 0|1:3:5:65,3 0/0:41:3
20 1110696 rs6040355 A G,T 67 PASS NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1|2:21:6:23,27 2|1:2:0:18,2 2/2:35:4
20 1230237 . T . 47 PASS NS=3;DP=13;AA=T GT:GQ:DP:HQ 0|0:54:7:56,60 0|0:48:4:51,51 0/0:61:2
20 1234567 microsat1 GTC G,GTCT 50 PASS NS=3;DP=9;AA=G GT:GQ:DP 0/1:35:4 0/2:17:2 1/1:40:3
```

- Header lines starting with #
- Data lines
 - encode genetic variants relative to a reference genome
 - specify position, reference allele (REF), alternative alleles (ALT) and genotype information for samples
 - genotypes encode presence or absence of alleles on sample's haplotypes
 - “[” : alleles phased (we know which haplotype carries which allele), “/” : alleles unphased (we don't know which haplotype carries which allele)

VCF representation of graph

- Reference genome as additional path
- line in VCF = top-level bubble in graph
- REF field encodes reference sequence
- ALT encodes all non-reference paths
- sample columns encode haplotypes paths

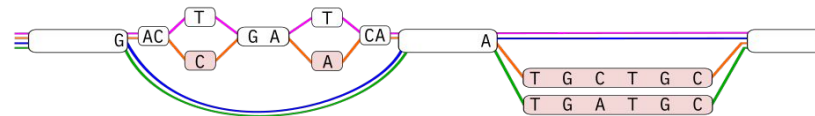


VCF records (multi-allelic representation)

CHROM	POS	REF	ALT	INFO	sample1	sample2
chr1	40	GACTGATCA	GACCGAACA, G	ID=chr1-43-SNP1:chr1-46-SNP2, chr1-40-DEL1	0 1	2 2
chr1	60	C	ATGCTGC, ATGATGC	ID=chr1-60-INS1, chr1-60-INS2	0 1	2 0

Representing nested variation

pangenome graph



Bubble VCF (PanGenie input)

- multi-allelic VCF
- INFO/ID annotations encoding nested variants
- each (nested) allele gets a unique ID

VCF records (multi-allelic representation)

CHROM	POS	REF	ALT	INFO	sample1	sample2
chr1	40	GACTGATCA	GACCGAACA, G	ID=chr1-43-SNP1:chr1-46-SNP2, chr1-40-DEL1	0 1	2 2
chr1	60	C	ATGCTGC, ATGATGC	ID=chr1-60-INS1, chr1-60-INS2	0 1	2 0

Callset VCF

- biallelic VCF
- each allele ID gets its own record
- makes comparison to other callsets possible

VCF records (bi-allelic representation)

CHROM	POS	REF	ALT	INFO	sample1	sample2
chr1	40	GACTGATCA	G	ID=chr1-40-DEL1	0 0	1 1
chr1	43	T	C	ID=chr1-43-SNP1	0 1	0 0
chr1	46	T	A	ID=chr1-46-SNP2	0 1	0 0
chr1	60	C	ATGCTGC	ID=chr1-60-INS1	0 1	0 0
chr1	60	C	ATGATGC	ID=chr1-60-INS2	0 0	1 0

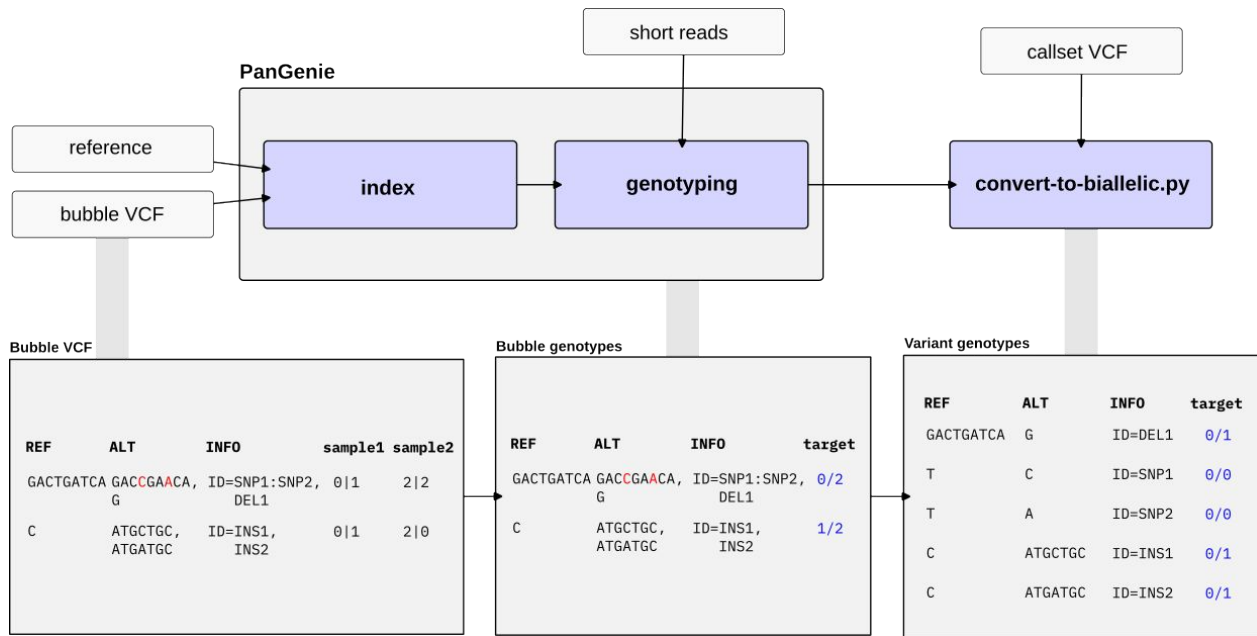
How to create such VCFs?

How to generate VCFs is not the focus of this workshop, but all information on this is available in the PanGenie Documentation (<https://pangenie.readthedocs.io>)

There we provide:

- pipelines to generate your own PanGenie input VCFs
- PanGenie-ready input VCFs for several HGSC and HPRC Minigraph-Cactus graphs

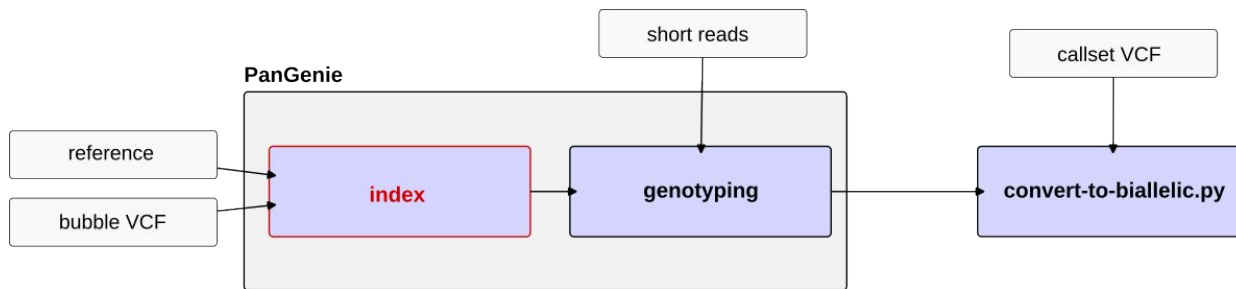
Focus today is to show how to use them with PanGenie



How to run PanGenie: indexing

- prepares data structures for genotyping
- determines unique k-mer sequences in the graph
- independent of sample data – needs to be **run only once**

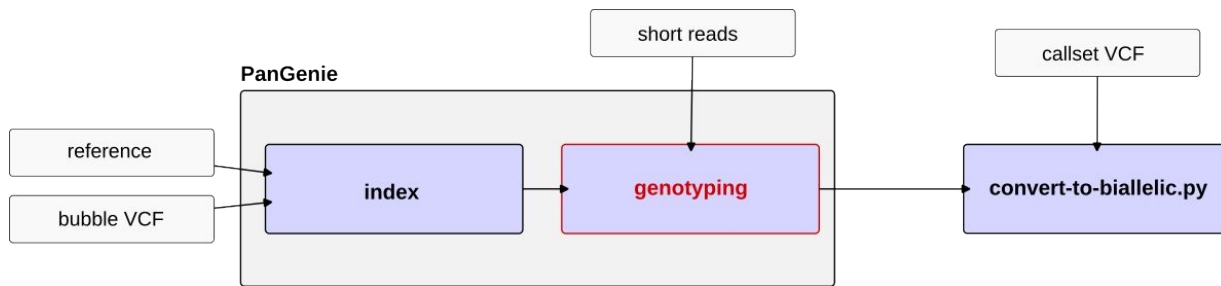
```
PanGenie-index -v <bubbles.vcf> -r <reference.fa> -t <number of threads> -o <index-prefix>
```



How to run PanGenie: genotyping

- count unique k-mers in the sample's sequencing data
- estimate average k-mer coverage
- genotype sample based on k-mer counts and haplotype paths in graph

```
PanGenie -f <index-prefix> -i <reads.fa/fq> -s <sample-name> -j <nr threads kmer-counting> -t <nr threads genotyping> -o <genotyping-prefix>
```



Convert bubble genotypes to variant genotypes

- Additional postprocessing step of PanGenie bubble genotypes
- Only possible with VCFs containing special annotations in INFO/ID field
- Converts bubble genotypes to genotypes for all (nested) variant alleles

```
cat <genotyping-prefix>_genotyping.vcf | python3 convert-to-biallelic.py <callset.vcf> > <genotyping-prefix>_genotyping_biallelic.vcf
```

